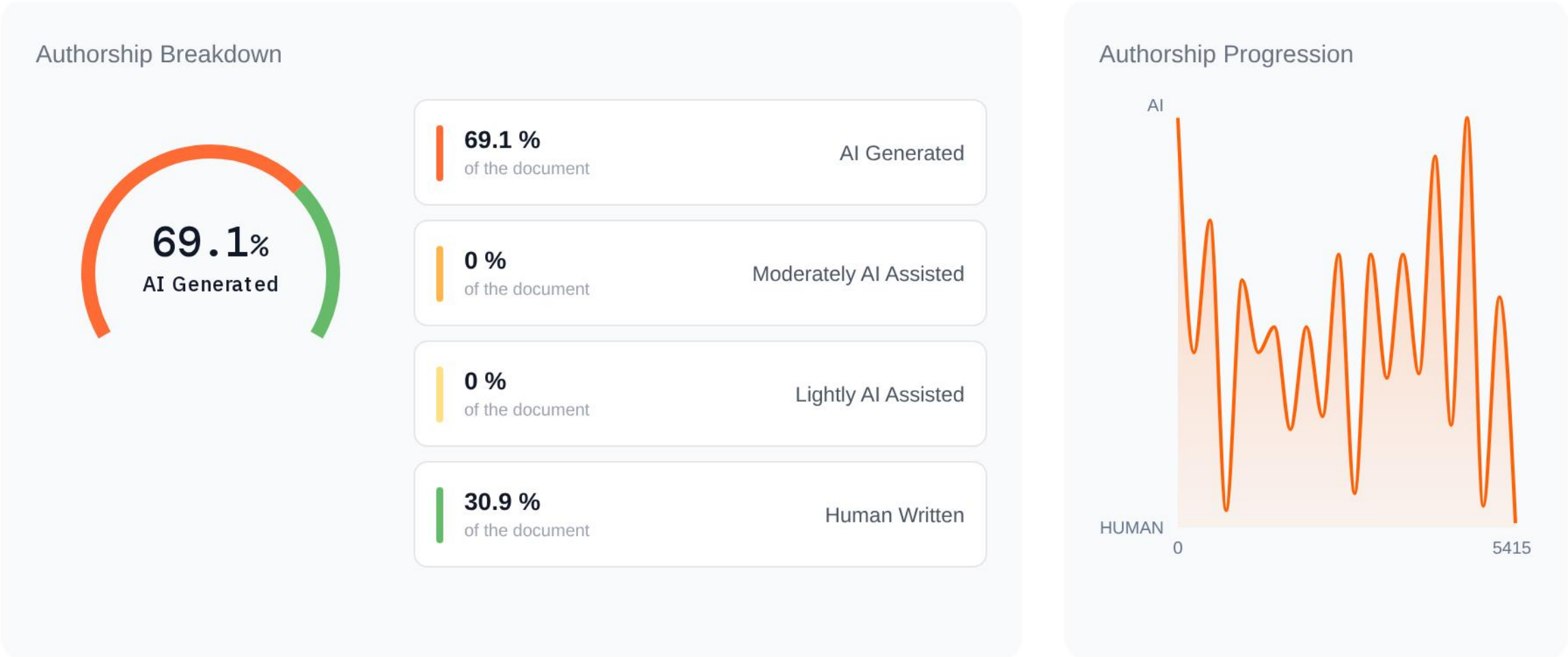


AI Detection Report for main.pdf

Aug 28, 2026 | 5,415 Words | Pangram 4.0

Summary

We believe that this text is a mix of AI and human-written content.



Analyzed Text

Words that are underlined indicate AI phrases, while the numbers represent how frequently they are likely to appear in AI writing. These elements are not used to determine the results but are displayed separately as patterns commonly found in AI-generated text.

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Energy-Aware Task Distribution in Mobile UAV Fog Networks Anonymous Author(s) Manuscript submitted for double-blind review Stage 2 is a permutation-invariant self-attention [13] encoder driving a PPO [14], [15] actor-critic that emits one (primary, backup) pair per task under feasibility masks. The elected head runs its own decision cost and carries its own control traffic, while the UAVs fly demand-following 3-D trajectories and pay Zeng-model [16] propulsion energy every control period. Contributions. (C1) A KL-gated, rank-reversal-free head election whose gate removes the ranking step on 97.6% of periods: at N=80 it runs 29–329× cheaper than three re- implemented baselines, switches heads zero times per 1000 tasks, and misses the fewest deadlines. (C2) An attention+PPO router, warm-started by behaviour cloning from a FODAS heuristic, which beats FODAS, ReLIEF, and headless random dispatch on task success rate and system energy per delivered task over an identical world, task stream and mobility trace. (C3) An evaluation that charges 3-D flight, propulsion energy and control-plane cost to whichever mechanism incurs them, against five re-implemented reference methods. II. RELATED WORK Abstract—Fleets of fog-computing unmanned aerial vehicles (UAVs) must continuously answer two coupled questions under battery, deadline, and reliability constraints: which drone co- ordinates the swarm, and which pair of drones executes each task. No prior fog-head election ties re-election to swarm drift, so a deployment gets either a stale head or a full re-ranking of every candidate on a schedule, at a cost none charge to the elected node; prior learned task schedulers ignore 3-D

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flight and price coordinator compute at a constant or not at all. We present a two-stage system. A Kullback–Leibler (KL)-gated multi-criteria election pipeline (MEREC → Kalman → KL gate → fixed-bound SPOTIS) skips 97.6% of full re-ranks. It feeds an attention-encoder PPO router that dispatches a (primary, backup) pair per task under demand-following 3-D mobility with Zeng-model propulsion energy. Against three re-implemented fog-head selectors (FU-Serve, 2DP-FHS, 3D-POS) at 80 UAVs, our selector costs 0.087 ms/period versus 2.50–28.62 ms (29–329× cheaper) and 0.087% versus 2.5–28.6% of head CPU, with zero head switches per 1000 tasks and the lowest deadline-miss rate (21.5% versus 23.3–49.2%). Against FODAS, ReLIEF, and headless random dispatch on an identical mobility trace and task stream, the routing stage improves task success rate and system energy per delivered task. All comparisons hold every mechanism constant except the one under test. Index Terms—UAV fog computing, task offloading, multi-criteria decision making, Kalman filtering, rank-reversal-free ranking, self-attention, proximal policy optimisation, energy-aware scheduling I. INTRODUCTION Fog computing [1] pushes computation to the network edge; a fog-UAV swarm puts it directly overhead [2]. Two questions then recur every control period. Which drone coordinates the swarm? And for each task, which drone runs it, with which second drone standing by? Neither answer holds still: batteries drain, queues grow, and the drones relocate to follow demand. Table I places this work among its neighbours. On the election side, FU-Serve [3] maximises a past-/current-condition composite over residual energy, connectivity degree and inter-UAV distance, re-running it on head-energy depletion. 2DP-FHS [4] and 3D-POS [5] rank candidates against a utopia point on a Pareto front of two and three objectives, over static health-care fog devices, and state no re-election trigger, nominating a failure backup instead; a FaaS-based selector [17] continues that line. None of the three carries reusable state between decisions, so cost tracks candidate count on every firing, and the scheduled alternative, LEACH-style rotation [18], pays it every round. Only FU-Serve models flight. Existing fog-head selectors (FU-Serve [3], 2DP-FHS [4], 3D-POS [5]) rescore the whole candidate set whenever they fire, and none ties firing to swarm drift. That leaves a stale head or a schedule that pays for rescoring out of the same CPU that runs the coordinator's own tasks, a cost none of them charge to the elected node. Learned schedulers have the opposite blind spot: FODAS [6] prices its broker at nothing, ReLIEF [7] at a constant, on nodes that never fly. Routing has been studied separately. FODAS [6] is a deadline-adaptive EDF-plus-cost heuristic, while ReLIEF [7] is a tabular Q-learner keyed on a system-wide reliability and workload pair rather than any per-candidate signal. Both halves have mature literatures of their own: rank-reversal-free MCDM [11], [12], [19] and UAV-MEC deep-RL offloading [20]–[23]. We join a cost-gated version of the first to a mobility- and control-cost-aware version of the second. III. SYSTEM MODEL We pair two mechanisms, both built to stay cheap as the swarm grows. Stage 1 elects a Master Fog Head: MEREC [8] sets the criterion weights, a Kalman filter [9] smooths them, a KL [10] gate suppresses the expensive ranking step whenever nothing has moved, and SPOTIS [11] ranks against fixed bounds, which makes it rank-reversal free by construction [12]. Fig. 1 shows the pipeline. Our swarm is $N=20$ heterogeneous UAVs, swept from 10 to 80, serving 10N IoT devices spread over a 5 km×5 km grid from a 4-hotspot Gaussian mixture whose centroids drift.

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One latent quality $q_j \in [0, 1]$ TABLE I POSITIONING AGAINST RELATED WORK. LEFT: FOG-HEAD ELECTION. RIGHT: TASK ROUTING. RR: RANKING IS RANK-REVERSAL FREE.

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3-D: 12 m/s for roughly 779 W, down in the induced-power dip, so a moving swarm can spend less than a stationary one. Every control period charges $P(V) \Delta t$ against residual energy E_{res} THE METHOD MODELS FLIGHT IN THREE DIMENSIONS. COST: THE METHOD CHARGES ITS OWN DECISION COMPUTE AND CONTROL TRAFFIC j , with $SOC_j = E_{res} j / E_{init,j}$. TO A NODE IN THE SYSTEM. EVERY ENTRY IS READ FROM THE CITED PAPER; RELIEF PRICES ITS BROKER AT A LOAD-INDEPENDENT CONSTANT.

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IV. STAGE 1: HEAD ELECTION A. Decision criteria Task routing 3-D Cost FODAS [6] no no ReLIEF [7] no const. Prop. (attn+PPO) yes yes Head election RR 3-D Cost FU-Serve [3] no yes no 2DP-FHS [4] no no no 3D-POS [5] no no no Prop.

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(KL+SPOTIS) yes yes yes Stage 1 ranks candidates on six criteria, recomputed from live node state every control period and signed (+) beneficial or (−) cost-type. 1) Residual energy $E_{res} j$ (+): Battery left after this period's flight, compute, radio, and idle draw: $[] E_{res} \Delta t$. (7) $P(V_j) + P_{proc} + P_{comm} + P_{idle} j \leftarrow E_{res} j - \text{fixes each UAV's hardware, interpolating its clock, MIPS, RAM, storage, battery and failure rates together. Tiers are never assigned: they emerge, ranked by computational efficiency } CE_{eff} j = fr_j / (c_j C_j^{106})$. Table IV carries the full parameterisation.

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A. Channel and delay 2) Memory efficiency ME_j (+): Free RAM and storage, normalised by a fixed constant rather than the live swarm, so a node's score never depends on which peers are alive: $() () () \propto M P + \beta M S_{tot,j} - M P_{occ,j} - M S_{occ,j} , 0, 1$.

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Air-to-ground links use a line-of-sight path-loss model [24] over the 3-D slant range $Z_{i,j}$ between the endpoints and node j 's cruise altitude: $ME_j = \text{clip } ME_{\text{norm}} \sqrt{\Delta x^2 + \Delta y^2 + z^2}$ $g_{i,j}(t) = \nu_1 Z_{i,j} - \nu_2 j$. (1) i, j , $Z_{i,j} = (8)$ Occupancies come from resident tasks, so ME falls under load.

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The uplink rate follows Shannon–Hartley, and transmission delay is the payload in bits over that rate: $()$, $T_u T_{Ri,j}(t) = \omega \log_2 i, j = 8 \cdot 1024$ si $1 + g_{i,j}(t) p$ N_0 (3) Reliability $R_0 i, j$ (+): Survival of independent compute (λ_j) and link (μ_j) failure processes over a nominal task's execution and transmission times: $T_{Ri,j}(t)$. (2) $()$ R_0 . (9) $i, j = \exp -\lambda_j li$ f avg T $Ri,j(t)$ $r_j - \mu_j$ $8 \cdot 1024$ si Execution and dual-queue EDF queuing delay for a node with mean clock f avg r_j and n_j resident tasks of cycle cost lk are $n_j - 1 \sum T e$, $T Q, avg$ $(n_j - k) lk$. (3) $j, i = li$ f avg $j = 1$ $n_j f$ avg r_j r_j $k=1$ 4) Controller delay D_j (-): How much slower than the fastest node j would service a controller job, given its clock and queued control work, with $T_{ctrl} j$ that projected service time: $()$ $()$.

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(10) $D_j = \max \min 0, T_{ctrl} j - T_{ctrl}$ The broker's own decision cost is a bounded per-period over- head, gated by the KL indicator $I_{trigger} \in \{0, 1\}$ (Section IV), and the total delay of task τ_i on node f_j is $T(k) B = T_{MER} + T_{Kal} + T_{KL} + I_{trig} T_{SPO} + T_{PPO}$, (4) This is a control-plane quantity, not a task delay: it asks who runs the coordinator fastest, which is the role being assigned.

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$D_i = T(k) j, i$. (5) $j + T e B + T u j, i + T Q, avg$ 5) Workload W_j (-): Cycles pending across both queues, backups included as committed work: B . Mobility and energy $\sum \sum lb$.

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(11) $W_j = l_p + p \in QP \quad b \in QB \quad j \in \{1, \dots, 6\}$ Energy coverage time ECT_j (+): How long residual energy lasts at current flight power and gated-controller duty g_j : Each UAV occupies a distinct slot on a 250 m standoff ring around its assigned hotspot (event-driven mode; random- waypoint and static modes are ablations) at a distinct altitude layer, so co-assigned UAVs never collide by construction. For propulsion we take the rotary-wing power curve of Zeng et al. [16], where blade-profile, induced and parasite terms combine as $(\cdot) (\cdot)^{1/2}$.

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(12) $ECT_j = E_{res} \frac{1}{1 + 3V^2} \frac{1}{1 + V^4} + \frac{1}{P(V)} = \frac{P_0}{P_i} \frac{U^2}{2V^2} \frac{0}{tip} \frac{j}{P(V_j)} + P_{proc} \frac{g_j}{4v^4} \frac{0}{-V^2} \frac{2d_0}{pAV} \frac{3}{3}$.

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(6) Hovering is just the $V=0$ point of that curve, and it is not the cheapest one. Our UAVs hover at 870.5 W yet cruise at ECT moves more slowly than Eres, so together they separate a momentarily-charged node from one that can sustain the role.  **PHYSICAL LAYER** — 3-D mobile fog over ground IoT elected Master Fog Head move swarm, charge $P(V)\Delta t$ recompute 3-D slant ranges 1 2 assignment 0.25 KB admission 80–150 m layers 0.25 KB task payload — data plane (IoT → executor, direct) 250 m standoff ring, hotspot-tracking hotspot 1 hotspot 4 200 IoT devices · 4 drifting hotspots · Poisson λ telemetry **DECISION + EXECUTION** — all on the elected head next control period Stage-1 elect (Fig. 2a) Stage-2 route (Fig. 2b) dual-queue EDF, backup held for a miss 3 4 5 Fig. 1. System architecture and the 100 ms control period. Task payloads fly IoT-to-executor directly on the data plane (solid); only small control messages route through the elected head (dashed), which also pays for its own Stage-1 and Stage-2 compute. Steps 3–5 are detailed in Fig. 2. B. Weighting, gating, and ranking A. State and permutation-invariant encoder MEREC weights each criterion by its removal effect: how much the swarm's aggregate assessment would change if that criterion were dropped, Each candidate contributes a feature vector, standardised by running EMA statistics ($\epsilon = 10^{-5}$) because attention scores are inner products and would otherwise be dominated by the largest-magnitude feature: $1/\sum_6 z_{j,k} = \Delta_{rem} j \quad c$.

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(13) $c=1 \quad \Delta_{rem} \quad x(k) \quad [\quad]^T, \quad \tilde{x}(k) \quad x(k) \quad SOC_j, \quad W_j, \quad R_0 \quad j, \quad TRB, f_j, \quad tier_j \quad j = j = j - \mu_{feat} \sigma_{feat} + \epsilon, \quad (\quad) \quad e(\ell) \quad e(\ell-1) \quad e(\ell-1), \quad j = LN \quad j + MHA(\ell)(\quad j ; \{e(\ell-1) \quad j' \}$
A Kalman filter smooths z_k into w_k across periods, trusting the smooth prior 100× more than any single measurement ($Q=10^{-4}I$, $\Sigma R=10^{-2}I$): $(\quad)$, $z_k - w_k | k-1 \quad w_k | k = w_k | k-1 + K_k$ (17) with $e(0) \quad j = W_{emb} \tilde{x}(k)$ (14) $(\quad) - 1$. $K_k = P_k | k-1 \quad P_k | k-1 + \Sigma R \quad j + b_{emb}$. Omitting positional encoding is what makes the encoder treat the swarm as a set, so one policy transfers across swarm sizes. B. Autoregressive policy with safety masks SPOTIS is the expensive ranking step in the pipeline, and claim C1 rests on how rarely it runs. It runs only when the weight

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vector has materially moved: $j \text{ } w_{j,k} \ln w_{j,k} w_{j,k-1} \geq \theta_{KL}=0.05$.

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(15) $DKL(w_k||w_{k-1}) = \sum$ Measured at $N=20$ across the arrival-rate sweep, that gate stays shut on 97.6% of periods. Stage 2 emits an ordered pair (f_p, f_b) , decoded autoregres- sively in two steps as in attention-based routing policies [25]. Each is a softmax over nodes, with infeasible choices driven to exactly zero probability by an additive $-\infty$ mask.

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Condi- tioning the backup on the primary's embedding h_{fp} links the two choices, and it cannot reuse the primary: $(\cdot), M(j) z(j) [h_j||c_k] p = -\infty$ if $SOC_j < 0.30$, $p = MLP_p$ When it does open, SPOTIS scores each candidate by its weighted distance to a fixed ideal solution profile rather than to the live candidate set, which is what makes the ranking rank-reversal free [11]: $(\cdot) z(j), M(j) [h_j||c_k||h_{fp}] b = MLP_b b = -\infty$ if $j = f_p$ or $SOC_j < 0.15$, $6 \sum, w_{j,k} D_x = |v_{x,j} - ISP_j| S_{max,j} - S_{min,j}$ (16) $j=1$ $f_{opt} = \arg \min_{x \in F} D_x$. (18) with $\pi_{\theta}(f_p|X_k) = \text{Softmax}(z_p + M_p)$, $\pi_{\theta}(f_b|X_k, f_p) = \text{Softmax}(z_b + M_b)$, and joint probability $\pi_{\theta}(A_k|X_k) = \pi_{\theta}(f_p|X_k) \pi_{\theta}(f_b|X_k, f_p)$.

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Decoding in two steps collapses an $O(N^2)$ action space into two linear ones while letting the backup depend on the primary. C. Reward Our reward pays for on-time delivery and charges for the ways a decision gets expensive. With $\Phi = \hat{W}$ a queue potential, the queue term is potential-based shaping and does not change the optimal policy: $+w_s \text{ slack}_k + w_{RR_i} - w_E \hat{E}_k$ Handovers are not free. Each charges a LEACH-style [18] 25 ms setup stall plus a 2 KB state broadcast to every UAV, so a switch only goes ahead when a lookahead test shows the candidate's projected delay advantage over the 10 s avail- ability horizon covers that cost. Safety overrides this: if the incumbent's SoC drops below the primary floor, the switch happens whatever the lookahead says. Elected-head control traffic (0.25 KB admission requests, 0.25 KB assignment mes- sages) travels the same physical channel as data; task payloads still move IoT-to-executor directly. (19) $r_k = R_{succ}$ or $R_{miss} \sim \text{outcome}(\cdot) + w_Q \gamma \Phi_{k+1} - \Phi_k - w_O 1[\text{overload}]$, V. STAGE 2: ATTENTION-PPO TASK ROUTING Placing a task means trading speed against battery against the odds of a node dying mid-task. Those weights shift too fast to freeze into a rule, so we learn them instead. overridden by $r_k = -M$ if the chosen primary would drop below its return-to-home floor, making a stranded UAV categorically worse than a missed deadline.  (a) Stage 1 — head election (per control period) Criteria X 2.4% SPOTIS fixed bounds head elected Kalman smooth w_k KL gate MEREC removal-effect weights (0.50 ms) (1.00 ms) (0.10 ms) $N \times 6$ Eres, ME, R, D, W, ECT $DKL \geq 0.05 \leq \theta$: 97.6% of periods, SPOTIS skipped (b) Stage 2 — task routing (per task, on that

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head) Attention encoder Actor: primary fp Actor: backup fb SoC $\geq 30\%$, $R \geq 0.95$ Features $N \times 18$, EMA-normalised dispatch backup held, sent on miss margin 25 ms 3 layers, 8 heads dmodel = 128 fb $\neq$ fp SoC $\geq 15\%$ Fig. 2. The two decision pipelines. (a) Stage-1 head election: the KL gate skips the SPOTIS re-rank on 97.6% of periods (measured, $N=20$). (b) Stage-2 routing: an attention encoder feeds an autoregressive actor that emits an ordered (primary, backup) pair under feasibility masks. D. PPO objective TABLE II STAGE-1 HEAD-TO-HEAD, $N=80$, $\lambda=720$ TASKS/S Training is behaviour cloning from FODAS, which avoids random exploration on a saturated network, followed by 300 PPO [14] episodes with GAE [15] advantages $\hat{A}_k$. Writing $rt(\theta)$ for the new-to-old policy ratio, formed in log space from the two stored component log-probabilities to avoid underflow, Selector ms/tk CPU% chg/1k miss% J/del. 2DP-FHS 18.068 18.07 13.47 49.2 1.352 3D-POS 28.618 28.62 9.76 38.4 1.219 FU-Serve 2.499 2.50 0.181 23.3 1.081 Proposed 0.087 0.087 0.000 21.5 1.065 [()] min , $LCLIP(\theta) = E_k rt \hat{A}_k, clip(rt, 1-\epsilon, 1+\epsilon) \hat{A}_k [] L(\theta, \phi) = E_k -LCLIP(\theta) + c1LV F(\phi) -c2H[\pi\theta]$. (20) Clipping caps how far one update can move any action's probability, so a single lucky sample cannot push the policy somewhere it cannot recover from. Evaluation is greedy with frozen weights. leave unspecified, re-rank whether or not anything changed, and pay one to two orders of magnitude for it. Panel (b) puts the same number in the unit an operator feels. At 80 UAVs Stage 1 takes 0.087% of the elected head's CPU under our pipeline; FU-Serve, 2DP-FHS and 3D-POS take 2.5%, 18.1% and 28.6%. Nearly a third of 3D-POS's coordinator goes on working out who the coordinator should be. Because primary and backup are distinct nodes they fail independently, so the task survives if either does, $R_i = 1 - (1 - R_{0i,fp})(1 - R_{0i,fb})$. Sending every task twice would waste the energy we are conserving, so the backup is reserved and dispatched only if the primary's result does not arrive in time. VI. EVALUATION Every comparison here is paired. We re-seed each configuration so the arms meet an identical world, task stream and mobility trace, leaving the mechanism under test as the only thing that differs. A. Stage-1 scalability None of that saving is bought with service quality, as panels (c) and (d) show. At $N=20$ our mean delay sits within noise of FU-Serve, 120.2 against 120.7 ms, and we claim no win there. What we do claim is cost and stability once the swarm grows. Past $N=40$ the Pareto selectors pull away, and by $N=80$ they miss 38–49% of deadlines where we miss 21.5%. Tails behave like the means: P95 of 354.9 ms against 358.4, 400.3 and 422.7 ms. Table II adds the failover test: we kill the elected head at $t=5$ s and watch. All four selectors recover inside a single control period. Our delay spike is the smallest, 49.7 ms against 53.5–55.2 ms, and we lose the fewest in-flight controller jobs. The lookahead test earns that: it keeps a well-provisioned successor waiting instead of churning between marginal candidates. B. Stage-2 task distribution What happens to a selector as the swarm it must rank gets bigger? Each configuration runs a fixed 10 s horizon. We scale the Poisson arrival rate alongside $N \in \{10, 20, 40, 80\}$ so offered load per UAV stays put: the ranking problem grows while each drone stays equally busy. Every condition repeats over 20 paired trials, all four selectors share one frozen Stage-2 actor, and intervals come from a paired bootstrap over 2000 resamples. We re-implemented the baselines from their source papers against the same node state and geometry. Now we hold Stage 1 fixed and vary the router instead. Six arms each run a 25,000-task episode at $N=20$ and $\lambda=180$ tasks/s. That loads the swarm to roughly 98% of its aggregate compute capacity, which is deliberate: near saturation is where a scheduling decision changes the outcome at all. Replication differs by arm. Three arms consume the learned policy, namely the proposed system and its static- and random- mobility ablations; those run over 3 training seeds $\times 11$ Panel (a) of Fig. 3 carries the result. Our selector's cost barely moves as the swarm grows: 0.058 ms at $N=10$, 0.087 ms at $N=80$. The KL gate is why. It lets Stage 1 skip the ranking step on almost every period,

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whereas the baselines, run here on the fixed schedule their source papers. Proposed FU-Serve 2DP-FHS 3D-POS 10 3 100 50 (a) (b) (c) (d) 100 ms budget 150 45 80 10 2 40 60 140 10 1 35 40 28.618% 18.068% 30 130 10 0 Mean delay (ms) 20 Deadline miss (%) Head CPU share (%) 2.499% 25 Selector cost (ms/period) 0 10 -1 120 20 0.087% 10 40 80 Fog UAVs N 10 40 80 Fog UAVs N 10 40 80 Fog UAVs N 10 40 80 Fog UAVs N Fig. 3. Stage-1 scalability. (a) Per-period selector cost, log scale. (b) The same cost as a share of the elected head's CPU. (c)–(d) Delivered performance over the same sweep. TABLE III STAGE-2 HEAD-TO-HEAD, $N=20$, $\lambda=180$ TASKS/S, MEAN $\pm$ 95% CI OVER SEEDS pounds. It drives the swarm past the state-of-charge feasibility floors sooner, and the nodes it strips from the candidate pool cost more late than proximity to demand ever bought early. Arm succ.% mean D (ms) J/deliv. Proposed 56.5 ± 1.9 140.5 ± 2.1 156.5 ± 4.4 static-mobility ablation 53.8 ± 2.0 143.8 ± 2.0 167.9 ± 5.1 random-mobility ablation 58.2 ± 2.1 139.9 ± 2.2 143.2 ± 4.4 FODAS [6] 33.9 ± 4.4 165.9 ± 3.6 268.2 ± 28.7 headless (random) 27.7 ± 2.7 170.3 ± 2.3 323.8 ± 27.2 ReLIEF [7] 50.3 ± 3.2 146.4 ± 3.4 176.1 ± 10.2 evaluation seeds, so 33 runs each. None of the three base- lines touches the trained checkpoint. Replicating those across training seeds would be pseudo-replication, not independent sampling, so they run over the same 11 evaluation seeds and no more: 11 runs each. All six route through the 6-criterion Stage-1 broker of Section IV, so one selector underpins both studies. Baseline rankings turn out to be sensitive to the causal control-plane cost model. FODAS's static 0.7-time/0.3-energy heuristic waits longer in queue here than ReLIEF's Q-learning policy, 112.9 against 86.3 ms, even though ReLIEF keys on a system-wide rather than per-candidate state [7], a documented risk at $4\times$ its original evaluation scale of 5–15 nodes. We report that reversal without claiming a confirmed mechanism behind it. Our three head-selection baselines also depart from their source papers in bounded, logged ways. None states a re-election cadence, so all run at Stage 1's; a coarser baseline schedule would trade per-period cost against head staleness. FU-Serve is further solved by exact enumeration rather than a GA, 2DP-FHS by $\alpha \cdot \text{norm}(\text{FDI}) - \beta \cdot \text{norm}(\text{FPI})$, which reproduces that paper's two unambiguous endpoints, and 3D- POS by 2/1 core counts per tier. All four Stage-1 service costs in Table II are declared simulation parameters taken from one-thread process_time_ns host timings, not hardware measurements. This is a simulation study throughout, with no hardware-in-the-loop validation. VII. CONCLUSION Table III gives the head-to-head and Fig. 4 the delay distribution behind it. Our learned router lands 56.5% of tasks on time, against 33.9% for FODAS and 27.7% for headless dispatch, and it does so on less system energy per delivered task: 156.5 J against 268.2 and 323.8 J. Of the two margins the energy one is wider: a missed task still burns whatever flight, radio and compute energy it consumed before anyone gave up on it. Read in isolation the absolute rates look modest. At $\approx 98\%$ offered load they look different: queueing delay climbs steeply there for any scheduler, and the comparison measures how gracefully each one degrades. Fig. 4(b) puts the difference in the queue term, the one a router can move. We joined a KL-gated, rank-reversal-free fog-head election to an attention-PPO router, under 3-D mobility and with control-plane cost charged where it falls. The gate is what keeps the election cheap as the swarm grows, and the router beats FODAS, ReLIEF and headless dispatch on delivery and on energy alike. Three things remain open: hardware-in-the- loop validation, energy-aware station-keeping, and joint Stage- 1/Stage-2 training.

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4(c) shows a trade-off in the mobility ablation rather than a clean win. Demand-following flight leads early: 76.8% of tasks delivered in the episode's first quarter against the random-waypoint ablation's 72.7%. By the third quarter that has inverted, 40.5% against 42.5%. That same panel shows why. Holding station on the standoff ring costs $\approx 4.6\%$ more propulsion energy than simply cruising, 2125 against 2028 kJ, because a station-keeping UAV sits above the induced-power dip of $P(V)$ that a cruising one rides.

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on mobile edge computing: The communication perspective," IEEE Communications Surveys & Tutorials, vol. 19, no. 4, pp. 2322–2358, 2017. Across a long episode with no recharge that penalty com- 250 1.0 249 235 (a) (c) (d) 60 389 400 0.8 (b) broker uplink queue exec 200 333 50 160 300 149 0.6 150 40 CDF 196 200 0.4 30 172 100 Mean SoC (%) Mean delay (ms) Energy / delivered (J) 20 0.2 100 50 event random static Proposed headless FODAS ReLIEF 10 0.0 0 0 ReLIEF ReLIEF FODAS FODAS headless headless Proposed Proposed 0 200 400 600 Delay (ms) 0 50 100 Time (s) Fig. 4. Stage-2 results. (a) End-to-end delay distribution across dispatch policies and (b) where that delay is spent. (c) Battery drain by mobility mode, the evidence for the demand-following crossover discussed in the text. (d) System energy per delivered task. [11] J. Dezert, A. Tchamova, D. Han, and J.-M. Tacnet, "The SPOTIS rank TABLE IV SIMULATION PARAMETERS. CITED VALUES ARE ADOPTED FROM THE REFERENCED WORK; THE REST ARE DESIGN CHOICES OF THIS STUDY. reversal free method for multi-criteria decision-making support," in 2020 IEEE 23rd International Conference on Information Fusion (FUSION), 2020. [12] X. Wang and E. Triantaphyllou, "Ranking irregularities when evaluating alternatives by using some ELECTRE methods," Omega, vol. 36, no. 1, pp. 45–63, 2008. [13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in Advances in Neural Information Processing Systems (NeurIPS), 2017. [14] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Prox- imal policy optimization algorithms," arXiv preprint arXiv:1707.06347, 2017. [15] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High- dimensional continuous control using generalized advantage estimation," arXiv preprint arXiv:1506.02438, 2015. [16] Y. Zeng, J. Xu, and R. Zhang, "Energy minimization for wireless communication with rotary-wing UAV," IEEE Transactions on Wireless Communications, vol. 18, no. 4, pp. 2329–2345, 2019. [17] R. Nanda, P. Sakthivel, R. K. Rath, and A. Hazra, "An efficient master head selection for multi-EEG to multi-fog IoT network using 6G-driven FaaS," Computer Communications, vol. 248, p. 108429, 2026. [18] W. B. Heinzelman, A. Chandrakasan, and H.

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Balakrishnan, "Energy- Topology 5 km×5 km; Nfog=20 (sweep 10– 80); Niot=10Nfog; 4-hotspot mix- ture, $\sigma=400$ m, drift 2 m/s Hardware clock 1.5–3.0 GHz [26]; 12–24k MIPS; RAM 2–8 GB; battery 162–234 kJ; λ_{fail} 0.025–0.005 s⁻¹ Workload Poisson $\lambda=180/s$ (sweep 60–300); Pareto size $\alpha=1.5$, [60,800] KB [2]; 1600–1800 cyc/B $\subset$ [200,1400] cyc/bit [26]; deadlines 150/550 ms Channel $f_c=2$ GHz; $B=30$ MHz, noise -174 dBm/Hz, IoT 150@34 + 50@45 dBm [26]; $u_2=2.3$ [24] Energy Phover=870.52 W, $P(V)$ curve [16]; cruise 12 m/s ($P\approx 779$ W); compute 10 W [2] Control $\Delta t=100$ ms; $\theta_{KL}=0.05$; handover 25 ms + 2 KB; dmodel=128, 8 heads, 3 layers; $\gamma=0.99$, $\epsilon=0.2$ [3] I.

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